

Embodied Intelligence Models: Progress, Mechanisms, and Relationships

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1. Scope and central findings

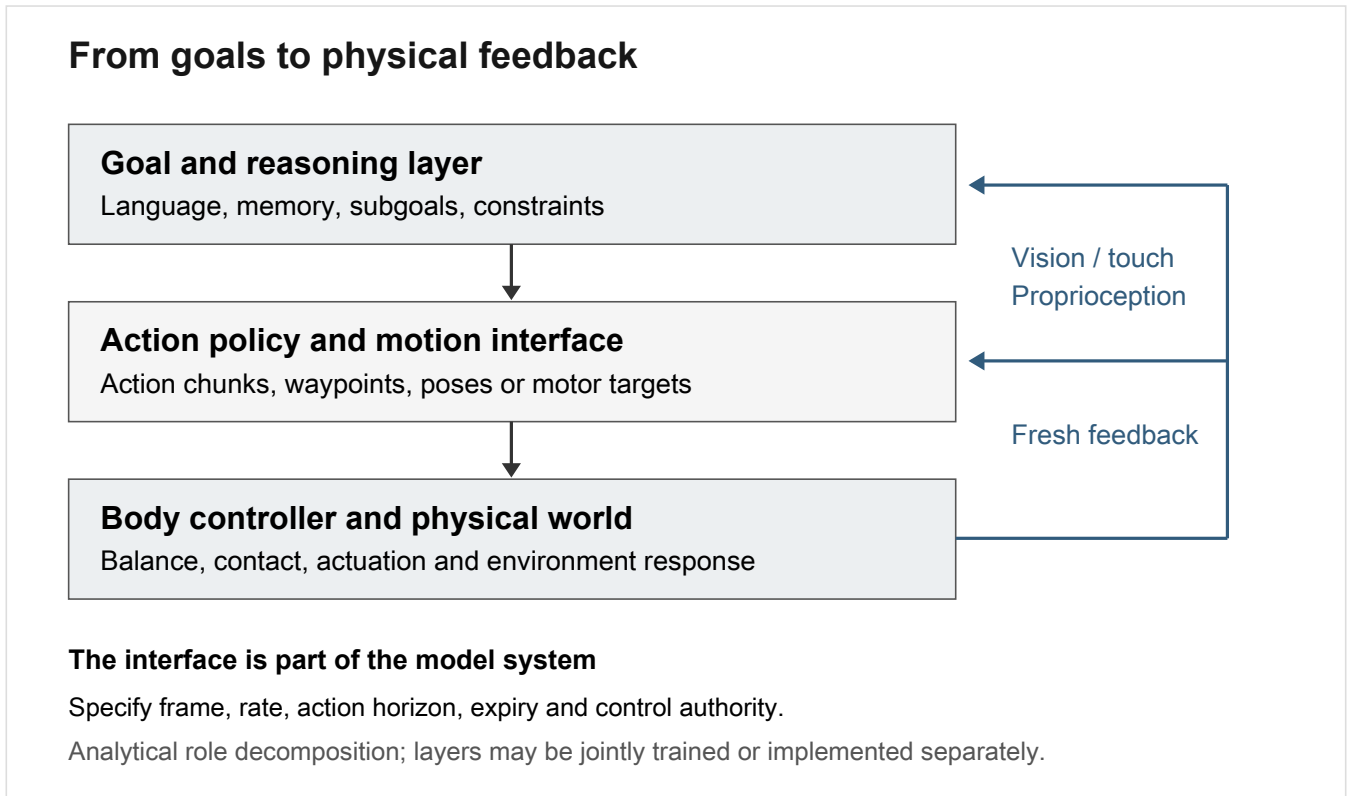
Progress in embodied intelligence connects semantic knowledge learned from internet data, action experience learned from physical interaction, and dynamics learned in simulation within a perception-action feedback loop. Parameter count and humanoid appearance alone miss decisive factors: data distributions, action interfaces, execution latency, and low-level control. This review focuses on robotic manipulation, navigation, and whole-body control, with driving world models as an adjacent reference. It does not attempt to survey all autonomous-driving perception, SLAM, or classical control.

Evidence was checked through **September 11, 2026**. The methodology is a technical scoping review: foundational work, original papers, author projects, official code, and product documentation are followed along research lines and updated with verified 2026 releases. This is not an exhaustive database-based systematic review with a measured retrieval recall, and no robot experiments were reproduced. Completeness refers to coverage of mechanisms, relationships, data, evaluation, engineering, products, and open problems within this scope. Paper results, vendor claims, and analytical judgments are distinguished; undisclosed architecture or training data are not inferred.

Five findings organize the report. First, **embodied intelligence is broader than a single VLA model**: representation, planning, action generation, prediction, and servo control are composable roles. Second, action generation increasingly uses chunks, diffusion, and flow matching, accompanied by asynchronous execution; these are design dimensions rather than one mandatory replacement chain. Third, world models need evidence of useful consequences, planning, or data generation; visual realism alone does not establish robot capability. Fourth, cross-embodiment data, domain adaptation, and feedback learning connect pretraining with deployment, while standardized file formats do not eliminate dynamics differences. Fifth, sustained reliability, contact feedback, and fair evaluation remain major gaps between demonstrations and operation. The model evidence and experimental conditions below support these findings.

2. A common decomposition and the technical foundations

Locate the model within the feedback loop



Closed-loop model roles

Let o_t include available images, depth, proprioception, and force or touch; let g denote the task, h_t history, and e the embodiment. An action policy can be written $\pi(a_{t:t+H} | o_t, h_t, g, e)$. A world model predicts $p(z_{t+1} | z_t, a_t, e)$, a high-level planner selects subgoals or skills, and a controller turns desired poses, velocities, or joint targets into execution. This is an analytical decomposition; it does not assume that every system implements separately named modules.

Role	Input and output	Representative lines	Essential verification
Visual/multimodal representation	Images or video to useful features	R3M, VC-1, V-JEPA	Improvements with downstream head and data held fixed
Semantic/hierarchical planning	Instruction and state to subgoals, skills, constraints	SayCan, PaLM-E, VoxPoser, Gemini ER	Feasibility, failure detection, replanning
Action policies and VLAs	Observations and task to actions or chunks	ACT, DP, RT, OpenVLA, pi, GR00T	Closed-loop success, latency, adaptation cost
World models	State and action to possible futures	Dreamer, UniSim, JEPA-AC, Cosmos	Effective conditioning, faithful consequences, planning gain
Whole-body and servo control	Proprioception and targets to frequent joint control	AMP, PHC, HumanPlus, SONIC, Helix S0	Balance, contact, disturbance recovery, hardware constraints

This separation explains apparently competing models. A visual encoder can serve different action heads, a VLA can call a whole-body controller, and a world model can produce training experience for an existing policy. Treating a language generator as a motor controller, or a video predictor as a deployable policy, obscures both interfaces and the evidence required.

Eight questions for every model

The decomposition asks: **What bottleneck is addressed? What is observed? How are modules connected? What action representation is produced? Which data and objectives train it? How are execution and feedback organized? Which design contributes the improvement? What limitations remain?** Parameter count is supplementary. For commercial systems without public architectures or training pipelines, only disclosed interfaces and system roles are described, with unknowns retained explicitly.

Even two systems described as a VLM plus an action expert may differ substantially. Is the VLM frozen? Does the expert read output features or layerwise KV states? Is training autoregressive, diffusion-based, or flow-matching? Are outputs normalized absolute joints or end-effector increments? How far ahead does the model predict, how much of that prediction is executed, and when does a new observation influence action? These choices determine adaptation requirements, throughput, and responsiveness.

Action representation determines transfer and control costs

Discrete action tokens quantize continuous values into classification targets that fit autoregressive language modeling, at a cost determined by sequence length, precision, and decoding. Direct continuous regression is simpler, but a conditional mean can fall between two individually feasible trajectories. Latent-variable and denoising approaches represent multimodal action distributions. Chunking predicts multiple timesteps together and improves coherence while creating a scheduling problem between observation and execution.

End-effector poses expose task geometry shared across arms but depend on inverse kinematics and reachability. Joint commands align more directly with hardware but bind the policy to joint order, offsets, and degrees of freedom. Torque outputs expose dynamics more directly to learning. Voxels, point clouds, and multiple views provide geometric inductive biases; force and tactile sensing reveal contact states hidden from vision. No representation dominates every task and embodiment; precision, data coverage, and the controller interface determine suitability.

Learning objectives and their relationships

Behavior cloning maximizes the conditional likelihood of demonstrated actions and provides useful initial skills, but distribution shift and missing recovery examples remain problematic. ACT-style latent-variable models and chunking improve trajectory modeling. Diffusion Policy learns to denoise action sequences, while flow matching learns a continuous velocity field transporting a noise distribution toward an action distribution, typically sampled by numerical integration. Both are conditional generators; neither automatically constitutes a model of the physical world.

World models learn transitions in latent or video space. Model-based reinforcement learning uses imagined trajectories to improve a policy during training, whereas model predictive control searches candidate actions during deployment. Both consider the future but at different stages. Reinforcement learning and experience-based post-training use rewards and corrections and can build on behavioral initialization. A rigid pretraining-to-cloning-to-RL sequence is not universal either.

Control rate and asynchronous execution require system-level accounting

Predicting H steps does not require waiting H steps before observing again: a system can execute a K -step prefix and replan, or infer the next chunk while buffered actions run. Asynchrony needs timestamp alignment, consistency with committed actions, removal of expired actions, and interruption under disturbances. With end-to-end latency L and control interval dt , merely covering the inference delay requires approximately $\text{ceil}(L/dt)$ buffered actions; excessive buffering makes control depend on stale observations. This is a scheduling constraint, not an empirical performance claim.

Xiaomi-Robotics-0's training prefixes and alignment, V-JEPA's planning budgets, and Helix/SONIC's hierarchical execution provide concrete examples later. Evaluation should record inference latency, observation age at actuation, jitter, control frequency, and success together. Actions generated per second are not the same as closed-loop robot responsiveness.

3. Manipulation policies and VLAs: from imitation to transferable robot foundation models

A VLA is more than a language model placed upstream of an arm. It aligns semantic information, visual observations, robot state and an action distribution through a trainable interface. Three questions organize the field: where knowledge comes from, how actions are represented and generated, and how predictions enter a real-time feedback loop. Parameter count, diffusion and dual-system branding each describe only part of that system.

Behavior cloning remains the underlying learning procedure for many robot foundation models. Diffusion, flow matching, action tokens and continuous regression are primarily alternative policy representations and objectives; using one does not establish that a model learns through autonomous reinforcement. RECAP makes a different change by introducing rewards, value estimation and policy-generated experience into training.^{[1][3][25]}

Action representations and architectural consequences

Route	Prediction and objective	Representative work	Benefit	Cost to inspect
Discrete controls	Action bins with classification/token prediction	RT-1, RT-2, original OpenVLA	Compatibility with transformer/VLM training	Resolution, sequence length and autoregressive latency
Continuous chunk regression	Multiple actions with L1/L2 supervision	OpenVLA-OFT	Parallel outputs and simple training	Conditional median/mean behavior for multimodal targets; language following
Continuous generative policy	Iterative generation of an action chunk from noise	Diffusion Policy, pi0, GR00T, SmoVLA	Continuous precision and multimodal distributions	Sampling cost and stale observations
Compressed action tokens	DCT, quantization and BPE on chunks	FAST, pi0-FAST	Less temporal redundancy	Compression error; training speed differs from feedback speed
Latent actions	Video-derived abstract action followed by a robot decoder	OpenDriveLab UniVLA	Learning without physical action labels	Loss of contact, reachability or control detail

The routes can coexist. pi0.5 uses discrete auxiliary action training and a continuous expert. Octo combines robot pretraining with a diffusion head. GR00T uses a module called a DiT but trains action generation with flow matching. Module names alone do not identify equivalent training algorithms.^{[6][8][15][19][21][31][47]}

Model decomposition: imitation and early foundation policies

ACT (2023). Small imitation errors accumulate over long tasks, while low-cost bimanual demonstrations vary. ACT takes multiview images and joint state into a transformer conditional VAE and predicts a joint-position chunk. Training combines action reconstruction with latent regularization; deployment uses a zero latent and combines overlapping predictions. Chunking improves fine manipulation from demonstrations, but the original method contains neither a general language foundation model nor guaranteed out-of-distribution force-control skills.^{[1][2]}

Diffusion Policy (2023). A scene can admit several valid actions. This policy models continuous action sequences through conditional denoising, with visual/state encodings supplying context. At deployment it samples a chunk, executes a prefix and observes again. It establishes a foundation for subsequent generative policies, but iterative sampling costs computation, and the original visuomotor policy is not automatically a web-pretrained VLA.^{[3][4]}

RT-1 (2022). To learn from diverse robot tasks at scale, RT-1 combines EfficientNet, language FiLM, TokenLearner and a transformer. It maps a short image history and instruction into discretized arm/base controls and a termination mode. The original dataset exceeds 130,000 episodes and 700 tasks; the policy runs at approximately 3 Hz. Its contribution is scalable multitask behavior cloning, with robot/task coverage still concentrated and language conditioning distinct from general reasoning.^{[5][6]}

RT-2 (2023). The question shifts from scaling robot data to transferring web semantics into control. PaLI-X or PaLM-E is co-fine-tuned on robot trajectories and vision-language tasks; binned actions become text tokens and are decoded back into controls. This supports semantic object selection and instruction generalization, but new semantics do not create unseen motor skills automatically. Autoregressive inference also adds latency. RT-2 is a research successor to RT-1, not continued training of RT-1 weights.^{[7][8]}

RT-X / Open X-Embodiment (2023). This is a cross-robot data and modeling framework with RT-1-X and RT-2-X variants, rather than one unique architecture. The initial report pools 22 embodiments, standardizes data interfaces and demonstrates positive transfer. Deployment must still restore target-specific action meaning, coordinates, units and frequency: a common storage schema does not remove dynamics differences.^{[9][10]}

Octo (2024). Octo pretrains a transformer diffusion policy on 800,000 OXE trajectories. Tasks can be given through language or goal images; modular inputs and action heads permit sensor/action adaptation. Inference samples a continuous chunk and replans after partial execution. Its role is a transferable robot-policy initialization, without requiring a giant language decoder; new platforms still need interface adaptation and fine-tuning.^{[11][12]}

OpenVLA (2024) and OpenVLA-OFT (2025). OpenVLA combines DINOv2/SigLIP visual features with Llama 2, learning discrete actions from approximately 970,000 demonstrations as a public 7B reference. OFT explicitly adapts that foundation with parallel decoding, action chunking, continuous outputs and L1 regression. It demonstrates the importance of output and fine-tuning design. However, L1 favors a conditional median; multimodal behavior and language grounding need separate evaluation. One benchmark does not prove regression universally superior to diffusion.^{[13][14][15][16]}

The pi family: continuous control, hierarchy, experience and memory

pi0 (2024). Multiview images, text, proprioception and noisy actions enter a PaliGemma 3B backbone with a roughly 300M action expert. Flow matching transforms noise into continuous chunks; the paper uses 50-action chunks and caches the observation prefix during sampling. Broad private/public robot pretraining is followed by high-quality task post-training. Generalist initialization and specialist dexterity results should therefore be distinguished.^{[17][18]}

FAST / pi0-FAST (2025). FAST is a representation: DCT, quantization and BPE turn temporally redundant action chunks into shorter discrete sequences. pi0-FAST is the autoregressive model using that representation; FAST itself is not a new robot architecture. Compression improves the utilization and training of high-frequency action data, but autoregressive latency remains, and compression ratio is not a feedback-speed metric.^{[19][20]}

pi0.5 and knowledge insulation (2025). The model retains the VLM/action-expert design and separates language-level subtasks from low-level continuous actions while mixing robot and nonrobot semantic data. Knowledge insulation blocks gradients from the action expert into the VLM, yet discrete-action and vision-language objectives still train the backbone. It is therefore not simply backbone freezing. The public openpi implementation supports only the pi0.5 flow head and should not be assumed to reproduce every training head in the paper.^{[21][22][20]}

pi0.6 and pi-star-0.6 / RECAP (2025). The pi0.6 model card specifies a Gemma 3 4B backbone and approximately 860M action expert, retaining pi0.5's hierarchy. RECAP adds offline reinforcement pretraining, demonstrations, corrective takeovers and reward-labeled autonomous rollouts. A value function provides advantages, and deployment conditions the policy on high advantage. A stronger supervised base and an experience-trained policy are different claims; throughput gains still depend on task, rewards and intervention infrastructure.^{[23][24][25]}

MEM / pi0.6-MEM (2026). To address occlusion, long workflows and repeated failures, MEM combines compressed short-term visual history with long-term language memory. Past attempts can then guide changed behavior, and completed subtasks remain available to the policy. The evidence concerns within-task memory and contextual adaptation, not indefinite lifelong or cross-user retention. State loss and incorrect summaries remain distinct failure modes.^[28]

pi0.7 (2026). Its paper explicitly retains the pi0.6 architecture and MEM while expanding conditioning: task/subtask, quality, speed, control mode and visual subgoals permit robot, human and suboptimal autonomous experience to be combined. At deployment a high-level semantic policy can provide subtasks and a lightweight BAGEL-based world model can supply visual goals to the continuous expert. The model shows skill composition and absorption of RL-specialist experience, but some new tasks use stepwise language coaching or high-level adaptation rather than one fully autonomous zero-shot prompt.^{[26][27]}

The inspected public openpi README lists pi0, pi0-FAST and pi0.5. The existence of pi0.6/pi0.7 papers and demonstrations does not establish release of their latest weights or complete data.^[20]

GR00T and compact VLAs: scale, embodiment interfaces and execution

SmoIVLA (2025). Designed for smaller compute budgets and community robots, it uses truncated SmoIVLM2 with an approximately 100M action expert, totaling roughly 450M parameters. Images, language and projected state condition flow-generated chunks; the VLM is frozen and the expert learns from LeRobot community and target data. Compact modeling and asynchronous execution lower the experimentation barrier. Synchronous, per-step-feedback and asynchronous evaluations nevertheless have different protocols and cannot be merged into one universal frequency or success rate.^{[29][30]}

GR00T N1 (2025). Eagle encodes vision and language; a cross-attending DiT combines these features with embodiment-specific state/action MLPs to generate continuous chunks. Flow training mixes real robots, simulation, generated trajectories and video-derived latent actions. Like pi0, it joins semantic representations with generative control, but its interfaces and data differ; this does not imply shared weights. The System 2 label does not require explicit textual planning.^[31]

N1.5 (2025). It improves and freezes the Eagle VLM, adds FLARE alignment with future latent features and expands data including DreamGen. Future-feature alignment is not full future-pixel generation. The authors explicitly note that new verbs supplied through synthetic trajectories are still trained skills, not unrestricted unseen-action generalization.^[32]

N1.6 (2025). Its technical page specifies an internal Cosmos-2B variant, a 32-layer DiT, unfrozen upper VLM layers and state-relative actions for most embodiments. Added data covers YAM, AgiBot, BEHAVIOR and Unitree G1. The page also identifies small-data relative-action accumulation, overfitting and language/OOD generalization as unresolved issues, highlighting action spaces and normalization as substantial engineering decisions.^[33]

N1.7 (2026). NVIDIA's engineering page describes Cosmos-Reason2-2B/Qwen3-VL, a roughly 3B checkpoint, around 32,000 hours of real/human data plus 8,000 simulated hours, and ONNX/TensorRT export. At inspection the official repository labels N1.7 GA. Its License section distinguishes Apache 2.0 code from weights under the NVIDIA Open Model License. A broader Apache 2.0 commercialization statement elsewhere in the README should not be read as applying the code license to weights; older early-access FAQ wording also requires version checking. Commercial availability is not demonstrated reliability in every field setting.^{[34][35]}

One source discrepancy is retained: the N1.6 technical page calls its backbone an internal Cosmos-2B variant, whereas the later N1.7 engineering article loosely describes its predecessor as Eagle. This survey follows the version-specific technical page rather than treating the broader wording as a precise architectural statement.^{[33][34]}

Gemini Robotics: action and embodied-reasoning branches

Gemini Robotics / ER (2025). The VLA maps images, language and robot state into movement; ER is a VLM for spatial understanding, task reasoning and tool use. The pair can form a hierarchical system, but ER alone is not a torque-producing controller. Reports establish Gemini-based robot capabilities, while full VLA weights and training corpora remain restricted. Undisclosed action losses and parameter counts must not be invented.^{[36][37]}

Gemini Robotics 1.5 / ER 1.5 (2025). Motion Transfer and interleaved reasoning/actions extend the VLA, while ER handles tools, plans and progress. Training covers ALOHA, bi-arm Franka, Apollo and web multimodal data. Its significance is cross-embodiment skill transfer and task-level orchestration. Some long-horizon Franka experiments add post-training, so the claim of operation without embodiment-specific post-training does not describe every result.^{[38][39]}

ER 1.6 (2026). Its model card identifies a Gemini 3.0 Flash foundation, targeting pointing, spatial relations, instrument reading and multiview success detection. Outputs remain text/spatial representations and tool calls. This is a reasoning-branch upgrade, not a new motor decoder. The authors note that single-view and multiview success detection use different examples, and instrument tests differ in agentic-vision support; these plots are not all matched-condition comparisons.^{[40][41]}

Robotics 2, ER 2 and On-Device 2 (2026). The July 30 release separates three roles: Robotics 2 for whole-body and dexterous action, ER 2 for understanding, planning and multi-robot coordination, and On-Device 2 for local execution and embodiment adaptation. ER 2's card identifies Gemini 3.5 Flash. On-Device 2 instead explicitly builds on Robotics 1.5 technology and on-device Gemma; it is not simply the full cloud model moved onto a device.^{[42][43][44]}

Robotics 2's published experiments still show significant multifinger and speed limitations. At launch ER was available through AI Studio, while VLA and On-Device models were for early-access partners. Local execution, API access and public weights are separate properties. Missing training losses, architectural details and full data inventories remain undisclosed.^{[42][44]}

Additional representative approaches and name disambiguation

RDT-1B (2024). A 1.2B robotics diffusion transformer combines language, images, state and control frequency for bimanual action generation. A physically interpretable unified action space supports heterogeneous pretraining, followed by bimanual fine-tuning. The contribution is physical meaning in the transfer interface, rather than merely padding vectors to equal length. Units, masks and coordinate conventions still require careful mapping.^{[45][46]}

OpenDriveLab UniVLA (2025). Video frame pairs and task text train a VQ-VAE latent-action model in DINO feature space. A VLA predicts these task-relevant latent actions, and target-robot supervision trains a lightweight decoder. This exploits human videos lacking physical robot-action labels, but abstract actions still require embodiment grounding and can omit fine contact or dynamics information.^{[47][48]}

BAAI UniVLA (2025). A separate same-named work starts from Emu3, representing vision, language and actions with discrete tokens and using world-model video post-training to improve policy learning. Authors, paper identifiers and mechanisms differ: task-centric latent actions versus unified multimodal generation. Its repository also displays CALVIN results with increased inference steps, requiring budget alignment before comparison.^{[49][50]}

X-VLA (2025). Florence-Large handles the main visual/language stream while auxiliary wrist views are encoded separately. Data-source/embodiment soft prompts and input/output projections adapt heterogeneous observations and controls; a standard transformer generates flow-matching action chunks. This separates shared capability from hardware-specific parameters, but soft prompts do not remove configuration or adaptation requirements for unseen hardware.^{[51][52]}

LingBot-VLA 2.0 (2026). A Qwen3-VL-based backbone, sparse MoE action expert and future semantic/depth distillation combine scaling with predictive control. Its approximately 60,000 curated hours comprise 50,000 robot hours across 20 configurations and 10,000 human egocentric hours. The 55-dimensional canonical interface includes four reserved slots and expands to head, waist, base and dexterous hands. The reported mixed-training GM-100 table evaluates nine tasks, and two long-horizon tests use 15 trials per setting; it does not establish that the entire benchmark or general OOD manipulation is solved.^{[53][54]}

Model relationships and defensible comparisons

Five relationship types matter: architecture/checkpoint inheritance, data reuse, auxiliary modules/objectives, training-procedure changes and system composition. OpenVLA to OFT is explicit adaptation; OXE to Octo/OpenVLA is a data relationship; pi0.6 to RECAP is RL post-training; MEM to pi0.7 is module inheritance; ER 2 and Robotics 2 are complementary components. Shared design among pi0, GR00T and SmoVLA does not prove checkpoint inheritance. The relationship catalog separately labels explicit evidence and analytical interpretation.^{[11][13][15][25][26][31][42]}

Execution comparisons must separate policy-call latency, actions generated per call, actual observation-refresh rate, low-level servo frequency and successful tasks per hour. RTC generates the next chunk while current actions execute and conditions on the committed prefix to maintain continuity. This is scheduling and conditioning, not additional semantic knowledge. Longer chunks can improve action-generation throughput while delaying response to new observations.^[55]

Matched evaluation requires the same robot, task and scene splits; training data and pretraining exposure; action space and control frequency; sampling steps, retries and adaptation protocol. LIBERO success, CALVIN chain length, GM-100 progress, long-task completion and hourly throughput measure different things and should not form one universal leaderboard. Cross-embodiment zero-shot claims must specify whether the novelty concerns objects, scenes, skills, bodies, or merely a skill-body combination.^{[15][32][38][50][53]}

4. Representations, hierarchical reasoning, and world models

These models occupy different positions in a robot system. A representation encoder describes the current state; a hierarchical planner selects a subgoal; a policy produces controls; a world model predicts the consequences of candidate controls. A model can occupy several positions, but real robot capability requires perception, decision, execution, and feedback together. This chapter distinguishes reported experiments, official product descriptions, and analytical conclusions. Versions were checked through 11 September 2026.

SayCan, PaLM-E, and EmbodiedGPT: connecting language reasoning to action

SayCan (2022) combines a language model's score for a skill's relevance with a value function's estimate of that skill's feasibility. It selects and executes a skill, then repeats. Its main training requirements lie in the skill policies and their value functions. The language model orchestrates abilities rather than generating joint commands. The original system receives environmental feedback primarily through current affordance values; its authors explicitly identify incomplete failure feedback as a limitation. The public tabletop simulation does not reproduce the full mobile manipulation stack.^[56]

PaLM-E (2023) embeds images and continuous robot state in the same dimensional space as language tokens. A decoder language model jointly trained on embodied and general multimodal tasks outputs textual plans for low-level policies, incorporating subsequent visual observations. The architectural change from SayCan is direct observation-conditioned reasoning. The 562B variant illustrates semantic transfer, not a 562B servo controller. Sensor grounding and downstream motor competence remain separate requirements.^[57]

EmbodiedGPT (2023) uses EgoCOT subgoal data derived from Ego4D to adapt a 7B language model with prefix tuning. Features extracted from planning queries inform downstream control. It explores a learned interface between visual evidence, subgoals, and action-relevant representations. Its benchmark improvements do not establish that verbal reasoning alone solves physical contact control.^[59]

VIMA and VoxPoser: task specification and geometric grounding

VIMA (2022 preprint; ICML 2023) represents tasks with interleaved text, object images, and demonstration frames. T5 encodes the prompt; a causal transformer with cross-attention conditions on that prompt and interaction history to produce manipulation actions. More than 600,000 expert trajectories and four generalization levels test changes in objects, compositions, and task templates. This object-centric policy is an important multimodal interface, but should not be conflated with every VLA that inherits an internet-scale language model. Its principal evidence is procedural tabletop simulation; real perception and action transfer require additional work.^[58]

VoxPoser (2023) uses language-generated programs and vision-language components to construct three-dimensional affordance and constraint maps. A motion planner converts those maps into end-effector trajectories. Re-evaluating cached programs with new visual observations enables receding-horizon adaptation. This makes spatial objectives inspectable, but introduces dependencies on segmentation, calibration, reachability, and dynamics assumptions. No additional training of the core composition does not mean no pretrained perception, controller, or integration effort.^[60]

R3M and VC-1: encoders are infrastructure

R3M (2022) learns visual features from Ego4D using temporal contrast, video-language alignment, and compactness regularization; downstream controllers can train on frozen features. **VC-1 (2023)** uses masked visual pretraining and CortexBench to study transfer across sensorimotor tasks, model scales, and data mixtures. Domain adaptation materially affects results. Neither encoder alone supplies actions, a long-horizon plan, or an action-conditioned transition model.^{[61][62]}

The analytical relationship to VLA is modular rather than a simple replacement lineage. An encoder can support a policy, reward estimator, retrieval system, or world model. Comparisons should hold controller architecture, demonstrations, cameras, and fine-tuning protocol fixed; otherwise apparent representation gains may come from other components.

PerAct, RVT, and 3D Diffuser Actor: geometry and generative actions

PerAct (2022) converts RGB-D observations into a voxel grid, combines voxel patches with language in a Perceiver, and classifies the next end-effector location and discretized orientation/gripper state. Aligning observations and actions in three-dimensional coordinates provides a useful prior for limited demonstrations. The costs include voxel resolution, memory, and calibration sensitivity; experiments cover RL Bench and a small real robot task collection.^[63]

RVT (2023) tackles the compute cost of explicit three-dimensional representations by rendering virtual views of the workspace and fusing them with cross-view attention. It predicts gripper poses while retaining geometric grounding. Relative to PerAct, the main change is how spatial information is computed. Reported speedups depend on the authors' implementation, hardware, and matched-performance target.^[64]

3D Diffuser Actor (2024) combines spatial tokens with diffusion over end-effector translations and rotations, using three-dimensional relative attention during denoising. Its target is a multimodal **action distribution**, not future video. RL Bench keyposes still require a low-level planner to reach them. The official implementation documents quaternion conventions and differences between evaluation protocols, illustrating why action coordinates and environment versions must accompany success-rate comparisons.^{[65][66]}

DreamerV3: learning consequences and improving behavior in imagination

DreamerV3 (2023 preprint; 2025 Nature publication) is a model-based reinforcement learning algorithm, not one pretrained checkpoint that controls every robot. It learns recurrent latent dynamics, reward, and continuation from interaction, trains an actor-critic on imagined trajectories, and incorporates new real experience. Normalization, transformations, and loss balancing improve cross-domain stability.^{[67][68]}

Control optimization uses compact latent states even if image reconstruction helps training. Unlike typical behavior-cloned VLA, policy improvement explicitly exploits predicted outcomes and rewards. One configuration across tasks does not mean one set of weights solves every task zero-shot. Exploration cost, reward specification, and accumulated model error remain obstacles in physical deployment.

UniSim: generated experience as a training environment

UniSim (2023 preprint; ICLR 2024) combines heterogeneous image, video, navigation, and robot datasets in a conditional video generator. High-level language or low-level controls specify the next interaction. Iterative generation provides experience for training planners and reinforcement learning policies; the paper demonstrates transfer in particular real tasks.^{[69][70]}

Its contribution is a path from video generation to action-conditioned training trajectories. For engineering use, one must verify that controls causally affect predicted outcomes, failures remain visible, and policies do not exploit simulator artifacts. This is Yang and colleagues' generative interaction model, not the separate autonomous-driving sensor simulator with the same name.

Genie 1, 2, and 3: interactive environments and their limits

Genie (2024) learns a video tokenizer, latent action model, and dynamics predictor from videos without ground-truth control labels. Inferred latent actions are not automatically robot motor commands. **Genie 2 (December 2024)** uses autoregressive latent diffusion conditioned on past frames and actions. **Genie 3 (August 2025)** demonstrates 720p, 24-frame-per-second interactive environments with consistency lasting minutes. Project Genie subsequently provided a restricted product prototype in 2026.^{[71][72][73][74]}

The series advances generated environments and agent training substrates. Genie 3's official limitations include restricted agent actions and difficult multi-agent interactions; prompt-triggered environmental events need not be actions performed by the agent. Navigating generated worlds or rendering a robot does not validate physical contact precision, torque control, or long-term safety. The checked official pages do not disclose Genie 3's complete training data and architecture, so Genie 2 internals should not be presented as verified Genie 3 internals.

GR-1 and GR-2: video pretraining for direct robot policies

GR-1 (2023) pretrains on video and adapts to robot trajectories. A GPT-style model consumes language, image history, and robot state while predicting both future images and actions. Visual prediction supplies temporal supervision, but action outputs can be executed without explicitly searching multiple imagined futures. **GR-2 (2024)** scales video pretraining to 38 million clips and more than 50 billion tokens, followed by joint video-action adaptation.^{[75][76][77]}

This family bridges video models and VLA: it adds temporal priors to semantic understanding and executable actions to generative modeling. Joint prediction alone does not establish model-predictive control at inference. GR-2's 97.7% concerns the 105-task Simple setting; unseen manipulation is substantially harder. It does not establish success in arbitrary homes or robot embodiments.^[91] Access to the complete pretraining corpus and reproducible training setup requires separate scrutiny.

V-JEPA 2, 2-AC, and 2.1: latent prediction and planning

V-JEPA 2 (2025) predicts masked video representations rather than reconstructing every pixel. **V-JEPA 2-AC** post-trains an action-conditioned predictor using fewer than 62 hours of DROID robot video and recorded actions. Image-goal features guide receding-horizon planning on real Franka arms. Zero-shot deployment means no environment-specific or task-specific training there; it does not mean arbitrary robot control emerges from action-free video alone.^{[78][79][81]}

V-JEPA 2.1 (March 2026) adds dense predictive supervision for visible and masked tokens, intermediate-layer supervision, joint image/video tokenization, and scaling. Its robot experiments reuse the 2-AC predictor architecture, public code, and planning protocol. After replacing the visual encoder, the action-conditioned predictor is retrained on DROID; this is not direct inheritance of trained 2-AC predictor weights. Table 6 shows cup-grasp success improving from 60% to 70% at approximately three seconds of planning. Reaching 80% uses a longer horizon and approximately fourteen seconds. The widely summarized twenty-percentage-point gain therefore includes more planning compute. Each skill has only ten evaluated tasks, and gripper timing remains a failure mode.^{[80][81]}

Analytically, latent prediction retains task-relevant changes while avoiding unnecessary visual detail; generative video preserves interpretable observations. These approaches can coexist: latent search can reduce planning cost and image generation can support inspection or synthetic data. The decisive question is whether the representation preserves information about contact, occlusion, and failure.

Cosmos Predict, Cosmos Policy, and Cosmos 3

Cosmos Predict 2.5 generates world video conditioned on text, images, or video; **Transfer 2.5** transforms scenes using controls such as depth or segmentation. These are initially environment and data-generation components, not robot controllers.^[82]

Cosmos Policy (January 2026) explicitly adds executable actions by fine-tuning Cosmos-Predict2-2B on target-platform demonstrations. It represents action chunks, future observations/proprioception, and values as latent frames within the existing diffusion process. It supports both direct action sampling and best-of-N planning over predicted outcomes and values. The paper separates direct-policy and planning evaluations; additional rollout data can improve the learned dynamics and value function. The extra search cost must accompany claims of higher success.^{[83][90]}

Cosmos 3 (released May 2026; checked through September) combines an autoregressive reasoning tower with a diffusion generation tower using a Mixture-of-Transformers architecture. The current official repository lists Super 64B, Nano 16B, Edge 4B, and DROID-adapted policy checkpoints. Edge has resolution and video-transfer restrictions; recommended embedded hardware does not make every inference mode real-time. The checked repository currently uses OpenMDW-1.1, which should not be replaced with a license description copied from an older Cosmos generation. Complete pretraining reproducibility, hardware latency, and transfer to unfamiliar embodiments require checkpoint-specific evaluation.^{[84][85]}

GAIA-2, 3, and 4: the autonomous-driving boundary

GAIA-2 (March 2025) generates controlled multi-camera futures; **GAIA-3 (December 2025)** emphasizes evaluation. **GAIA-4 (August 2026)** places the AI Driver in a generated sensor loop, including camera and radar output. In world-on-rails mode, other road users retain logged trajectories for controlled counterfactual comparison; an optional reactive-agent mode allows selected participants to respond to the ego vehicle.^{[86][87][88]}

These are driving data and validation systems, not direct evidence for general manipulation. The transferable lesson is to assess visual fidelity, closed-loop behavioral consistency, and agreement on outcomes separately. Safety implications depend on scenario coverage, sensor accuracy, behavioral assumptions, and counterfactual fidelity. The checked GAIA-4 publication does not provide enough architecture and training-data detail for full reproduction, so it remains official system-level evidence.

Relationships and architecture choices

Relationship	Interpretation	Evidence category
SayCan → PaLM-E	Skill relevance/affordance interfaces versus observation-conditioned language planning	Analytical comparison; no weight lineage asserted
R3M / VC-1 ↔ policies	Reusable perception with separately trained control heads	Modular complement
PerAct → RVT	Similar spatial action grounding with a different geometric compute path	Explicit paper comparison
Geometry + diffusion → 3D Diffuser Actor	Spatial attention conditions action denoising	Explicit method combination
GR-1 → GR-2	Scaling video pretraining and joint video-action modeling	Family/method progression
V-JEPA 2 → 2-AC; 2.1 encoder + 2-AC framework	Encoder-family extension; predictor architecture and planning reuse with retraining	Branching extension/framework reuse, not weight inheritance
Cosmos Predict2 → Cosmos Policy	Video initialization adapted to actions, future state, and value	Explicit initialization
Earlier Cosmos lines → Cosmos 3	Joint reasoning and generation architecture	Official evolution; not a claim of direct weight inheritance
Genie / UniSim / GAIA ↔ policies	Generated training or evaluation environments	System division of labor
Dreamer ↔ VLA	Outcome-based policy improvement can complement demonstration-based initialization	Analytical combination

Choose the missing capability before choosing a model family. Skill composition suggests hierarchical planning; inadequate spatial precision suggests better geometry and action representations; abundant action-free video suggests representation or video pretraining; comparing alternative action consequences motivates explicit world-model planning. **MoLe-VLA is a dynamic layer-skipping efficiency method, not a world model.**^[89] Tasks involving tactile contact, persistent occlusion, human collaboration, and long-term state require additional feedback and state estimation beyond these visual approaches.

5. Whole-body control and navigation: turning model outputs into sustained action

Embodied systems span at least three problems: choosing a goal, generating executable short-term actions, and remaining stable under contact and disturbance. A model that can describe loading a dishwasher does not thereby provide balance, obstacle avoidance or grasp control. Relationships between models should first be located within this hierarchy; advances in motor control often complement a VLA rather than replace it.^{[99][102][110]}

From demonstrations to reusable motion priors

DeepMimic (2018) combines motion imitation with task rewards. Body state, phase and a goal condition a policy that produces joint targets and learns through physical feedback. Its contribution is to optimize resemblance and task achievement together. The original demonstrations concern simulated characters, including robot-shaped models; they do not establish deployment on physical Atlas hardware.^[92]

AMP (2021) and ASE (2022) shift emphasis from tracking a selected clip toward learning a distribution of movement. AMP uses a discriminator to provide a motion-style reward, reducing manual clip selection. ASE adds a latent skill interface so a higher-level policy can reuse the learned repertoire. ASE is an explicit methodological extension; the comparison with DeepMimic is an analytical change in problem formulation, not inherited weights. Natural movement still does not imply task correctness, and abrupt skill switching can create poor transitions.^{[93][94]}

PHC (2023) asks whether tracking can continue after noisy inputs or falls. Progressive multiplicative control adds capacity for difficult motions and recovery, making reset avoidance part of the controller. Its central evidence concerns simulated avatars. Transferring this result to real actuators, sensors and physical human interaction remains a separate challenge.^{[95][96]}

Humanoid Transformer (2023) instead uses observation-action history to autoregressively predict control actions, trained with model-free RL in randomized simulation. Behavior adapts through context without changing weights. This is contextual adaptation to execution dynamics, not the same capability as learning a new manipulation task from a demonstration video. Evaluation chiefly covers locomotion, terrain and disturbances.^[97]

From human motion to autonomous-task data

H2O and OmniH2O (2024) retarget human motion, filter infeasible examples using privileged imitation, and train deployable tracking policies. OmniH2O explicitly follows H2O and makes kinematic pose a common interface for RGB, VR, generated motion and learned autonomous policies. A privileged teacher must be distilled into a student whose inputs exist at deployment. Changing the goal generator need not entail rebuilding the physical controller. Teleoperated trials must nevertheless be counted separately from autonomous trials.^{[98][99]}

HumanPlus (2024) closes a data-collection loop. A low-level policy trained from 40 hours of human motion enables camera-based humanoid shadowing; resulting real-robot demonstrations support egocentric behavior cloning. A customized 33-DoF humanoid demonstrates multiple tasks. The useful sequence is human motion prior → practical teleoperation → robot task data → visual policy. Success on the reported small task set is not a measure of general competence across homes.^[100]

MaskedMimic (2024) represents heterogeneous commands as incomplete motion descriptions. Partial joints, paths, scenes or text constrain a motion-inpainting problem. Training obtains a fully constrained tracking teacher and then a conditional-VAE policy with randomized missing information. This admits multiple valid movements for an underspecified goal, rather than forcing one mean solution. The main evidence remains physics-based animation rather than a ready-made real-humanoid controller.^[101]

SONIC: an explicit interface between VLA and motor control

SONIC (2025 preprint; 2026 publication and engineering updates) encodes human, robot and hybrid reference motions into a shared quantized token. A common decoder outputs desired joint positions for PD controllers. PPO tracking, auxiliary reconstruction and domain randomization train the interface; a runtime kinematic planner can convert intent into short reference movements. This connects VR, human video and VLA commands to one execution policy.^{[102][123]}

The original paper fine-tunes GR00T N1.5 on 300 teleoperated demonstrations to produce head/wrist poses, waist height and base-motion commands, then executes those commands through SONIC. Its 20-trial, single-task experiment is explicitly a compatibility proof of concept, not a broad autonomy score. The inspected official repository now supplies an N1.7 workflow, training/deployment code and SONIC v1.1. Released controllers run at 50 Hz. Reference lookahead, neural inference duration and end-to-end teleoperation latency are different measurements.^{[102][103]}

HoloMotion provides a 2026 extension-reading example of a reference-conditioned MoE transformer and motion-data-to-deployment pipeline. It reinforces the relevance of model/data scaling in whole-body control, but a foundation-model label alone does not establish unseen-contact robustness, arbitrary-terrain competence or certified safety.^[124]

Navigation: local policies, global planning and spatial memory

ViNT (2023) learns transferable navigation from visual history and a goal image. Its transformer outputs local waypoints and a goal-distance estimate, while alternative goals can be encoded into the same interface. Long-distance behavior results from local policy, subgoal generation, topological representation and planning heuristics working together; it is not a single network directly producing every motor command.^{[104][105]}

NoMaD (2023 preprint / 2024 conference) explicitly reuses ViNT's encoder. A diffusion action decoder represents alternative routes, while masking the goal lets one policy support both exploration and directed navigation. This avoids simply averaging incompatible paths around an obstacle. Sampling, path selection, local control and new observations remain part of the deployed loop. Low collision rates in an experiment are neither a formal collision guarantee nor a humanoid balance result.^[106]

GOAT (2023) addresses object and place memory across missions. Observations update an instance-aware semantic map; category, image or language goals identify a target; the system explores for unknown objects and reuses remembered evidence for known ones. It is a modular system, and adaptation primarily comes from updating memory. **GOAT-Bench (2024)** is the distinct lifelong-navigation benchmark. A local navigation policy, semantic memory and VLA could be complementary, but that architectural inference should not be presented as an implemented model lineage.^{[107][108]}

Commercial hierarchies and contextual adaptation

Figure Helix → **Helix 02** is an explicit version extension. The original model emphasizes upper-body control. Helix 02 adds S0: S2 supplies semantic goals; S1 produces full-body joint targets at 200 Hz; S0 handles fast balance/contact execution at 1 kHz. Figure discloses over 1,000 hours of retargeted human motion and simulated RL for S0, alongside palm vision and fingertip touch. Its four-minute room-scale task is a vendor demonstration, not a multi-environment reliability study. A unified system still contains distinct levels and update rates.^{[109][110]}

Skild Brain → **S1** combines cross-embodiment learning with task adaptation. S1 describes episodic demonstration-conditioned pretraining and infers intent, progress and actions from a task video plus current context, without task-specific weight updates in the reported setting. The inspected article does not disclose a complete network, action parameterization or reproducible training recipe. Those fields remain unknown; the phrase general-purpose brain does not justify inventing a transformer or diffusion architecture. The public scaling study uses cumulative per-step scoring with human failure recovery; that score should not be equated with uninterrupted end-to-end autonomous task completion.^{[111][112]}

Generalist GEN-1 → **GEN-1.5** offers a useful contrast. GEN-1 emphasizes reliability after selected-task adaptation; GEN-1.5 places sensorimotor demonstrations in a 30-second context and produces 100-Hz trajectories. Generalist says its contextual ability emerged without a dedicated ICL objective, whereas Skild describes demonstration-conditioned training. Inputs, training and test distributions differ, so a single success-rate ranking is inappropriate. Generalist also explicitly describes current tasks as short and its contextual policies as more brittle than fine-tuned ones.^{[125][126]}

Product progress and the strength of evidence

Reproducible models, development platforms, vendor demonstrations, customer-confirmed pilots and sustained commercial operation are different stages. Each deployment claim should name the robot, model version, human assistance and metric denominator.^{[114][117][118][128]}

Organization	Technology/product	Verified progress	Evidence boundary
Figure	Helix 02 / Figure 03	BMW confirms the earlier Figure 02 pilot and new Figure 03 sequencing project; Figure publishes onsite demonstrations	Historical Figure 02 production statistics do not transfer to Figure 03 or Helix 02. ^{[127][128]}
Skild AI	Cross-embodiment Brain / S1	Company reports paying deployments and new S1 pilots	Commercial totals are vendor statements across products; deploying, planned and piloting are distinct. ^[113]
Physical Intelligence	π models integrated by partners	Weave laundry and Ultra warehouse customer-site accounts	Deployed versions differ from π 0.7 research; remote intervention remains part of operation. ^{[27][114]}
NVIDIA	GR00T, SONIC, simulation and onboard stack	Released models/code/workflows and announced humanoid reference hardware	Hardware availability is stated as late 2026, not delivered by the cutoff. ^{[103][33][115]}
Google DeepMind	Robotics 2 / ER 2 / On-Device 2	July 2026 whole-body, multi-embodiment and local-execution releases	Reasoning API access, trusted testing and open control weights are different access categories. ^[42]
Agility Robotics	Digit + Arc industrial service	GXO confirms a commercial RaaS deployment; vendor reports cumulative tote milestones	Counts omit denominators for failures, intervention and downtime; do not infer a disclosed VLA. ^{[118][119]}
1X	NEO / Redwood / 1XWM	Ordering and 2026 delivery plans; world-model technical disclosure	Basic autonomy and scheduled Expert Mode are explicit; orders are not proof of a delivered autonomous fleet. ^{[116][117]}
Boston Dynamics	Electric Atlas and learned behaviors	Official manufacturing, industrial pilot history and scheduled deployment	Hardware productization does not establish model reproducibility or general long-term autonomy. ^[120]
Unitree	G1 platform / UnifoLM-VLA	Official code, checkpoints and datasets	Hardware sales, model release and autonomous commercial usefulness need separate evidence. ^[122]
Galbot	G1/S1 and AstraBrain	Official WAM/WBC/dexterity and industry-application positioning	First/largest/scaled-deployment language remains vendor positioning rather than a technical ranking. ^[121]
Generalist	GEN-1 / GEN-1.5	Early enterprise access and real-robot adaptation/context studies	High reliability on adapted tasks and one-shot performance on unfamiliar tasks are different conditions. ^{[125][126]}

Reading model relationships and engineering implications

Explicit relationships include AMP→ASE, H2O→OmniH2O, ViNT→NoMaD, Helix→Helix 02 and the released GR00T-SONIC composition. PHC versus SONIC, or SONIC versus Helix 02, can be compared as approaches to placing broad motor competence below goals, but there is no claim of shared weights or code. The relation catalog marks explicit and analytical separately.^{[94][99][102][103][106][110]}

An engineering interface should specify coordinate frame, update rate, action hold duration, command expiry and control authority. Semantic decisions can tolerate slower updates; balance/contact feedback cannot wait for a remote reasoning round trip. Human references also require reachability and dynamic-feasibility handling. These are engineering deductions from the compared systems, not claims that every product uses the same implementation.^{[98][102][110]}

Unresolved issues include stability under unfamiliar contact and payloads; preservation of mechanical information across embodiments; automatic detection and recovery over minutes; and evaluation that jointly reports success, throughput, human time and failure cost. More natural motion, more realistic video or a larger network alone cannot settle these questions.^{[100][102][114][116]}

6. Additional decompositions: data, feedback, and execution

GO-1: a latent plan between semantics and continuous action

GO-1 consumes multiple camera views, instructions, and robot state. Its ViLLA architecture combines a vision-language backbone, a latent-action planner, and a diffusion action expert. Discrete latent actions are learned from sources including human video; a planner predicts them from the current context, and the action expert conditions on this representation to generate continuous chunks. Latent-action prediction and robot action denoising have distinct objectives. GO-1 Air removes the planner, providing a public entry point for investigating whether the intermediate planning layer is useful.^{[132][133]}

This is a combination of semantic transfer, latent-motion transfer, and continuous control, rather than the inevitable successor to every VLA. Latent actions can use video without robot-command labels but still need grounding into executable commands through robot data. When architecture and data scale change together, gains cannot be attributed solely to the planner; matching task, data, and compute while comparing GO-1 with Air is more informative. AgiBot World’s collection-robot count describes hardware instances, not the same number of distinct embodiments.^{[132][133]}

SERL and HIL-SERL: learning from corrective experience

SERL integrates training, experience, and robot interfaces for sample-efficient visual reinforcement learning; it is not one pretrained weight set covering all tasks. HIL-SERL introduces a concrete feedback workflow within this line: collect positive and negative states for a reward classifier, initialize a demonstration buffer, execute online, intervene when needed, and use the resulting experience in actor-critic learning. The learning signal includes visual observations, state, sparse success signals, and human corrections.^{[142][143][153]}

Relative to imitation, the additional information concerns recovery after deviations and whether actions actually succeed. Relative to generalist VLAs, the method is closer to precise post-training on a bounded task. Reported learning speed and high success cannot be separated from reward labels, interventions, resets, and hardware. Engineering comparisons should include cumulative human minutes, reset count, and usable interaction per hour, rather than gradient-update time alone.^{[142][153]}

MolmoAct2: an open embodied-reasoning backbone with continuous control

MolmoAct2 builds on the Molmo2-ER embodied-reasoning vision-language backbone and adds robot state and a flow-matching continuous action expert. Images, language, and state condition action chunks for closed-loop manipulation. Public assets include intermediate foundation checkpoints, selected robot-adapted policies, and relevant datasets, enabling analysis of how action generation uses reasoning representations. Producing explanatory text is insufficient by itself: the action module must still learn an appropriate command distribution from robot trajectories.^{[144][145]}

The released checkpoints make the training path more explicit. Molmo2-ER is the starting vision-language backbone; MolmoAct2-Pretrain is a discrete autoregressive VLA before the continuous action expert is attached; the final MolmoAct2 connects a flow-matching expert. MolmoAct2-Think adds depth-token reasoning and should be identified as a separate variant, rather than attributing that mechanism to every deployed MolmoAct2 policy. This separates two research questions: how discrete action pretraining shapes the backbone, and whether depth reasoning improves continuous control. Their contributions still require ablations with a fixed robot, data mixture, and inference budget; releasing intermediate checkpoints does not by itself establish causal attribution.^{[144][145]}

Its relationship to GR00T, PI, and Xiaomi concerns a modular division between visual-language understanding and continuous generation, not shared weights. Backbones, training mixtures, alignment objectives, and deployment protocols differ. Staged checkpoints facilitate ablations, while improvements in contact, unfamiliar objects, or long-horizon recovery require matched evaluation. Foundation checkpoints and embodiment-adapted deployment policies should be treated as different assets.^[144]

Xiaomi-Robotics-0: training for the execution schedule

Xiaomi-Robotics-0 combines Qwen3-VL-4B-Instruct with an action DiT, totaling 4.7B parameters according to the official project. A first stage develops action-relevant VLM features using Choice Policies and mixed vision-language data. A second freezes the VLM and trains a flow-matching expert conditioned on its KV states and proprioception. Post-training introduces action prefixes and attention/loss design so that new chunks agree with committed actions while remaining responsive to observations.^[151]

The public deployment implementation returns 30-step chunks. Asynchronous execution predicts the next chunk while current actions run and removes actions that have expired during inference. This mode requires a correspondingly trained checkpoint; attaching any synchronous policy to an asynchronous queue is not equivalent. The broader lesson is that **latency compensation changes training conditions and distributions**, rather than being an isolated runtime patch. Strong reported real-task performance also uses substantial specialized data, so task proficiency, few-shot adaptation, and zero-shot generalization remain distinct claims.^{[151][152]}

1XWM: grounding generated video into action

1XWM generates a future from a prompt and starting frame, then uses inverse dynamics to extract NEO commands. The January 2026 disclosure describes a 14B video backbone adapted with egocentric human and robot data, and a Depth Anything encoder with a flow-matching head for the inverse model. Video priors therefore still require an action-grounding interface.^[116]

That version needs approximately 11 seconds of multi-GPU inference for five seconds of video plus one second to extract actions; longer-horizon closed-loop replanning remains future work. This separates the demonstration from reactive, on-device, sustained household autonomy. These limitations apply to the disclosed version, without assumptions about unpublished improvements.^[116]

7. Data and evaluation: establishing actual progress

What the data assets contribute

Asset or interface	Verified representative scale or property	Relevance
Open X-Embodiment	22 robots in the original paper; standardized records	Cross-embodiment sharing; actual mixtures differ by model ^[9]
BridgeData V2	60,096 trajectories in 24 environments, including teleoperation and scripted runs	Object and scene diversity; overlaps with OXE mixtures ^[148]
DROID	76k trajectories, 350 hours, 564 scenes	Distributed real-world collection; calibration/data revisions matter ^[129]
RoboMIND	Original paper: 107k trajectories, 479 tasks, 96 object classes	Multiple embodiments collected on a unified platform ^[131]
AgiBot World	Beta lists 1,003,672 trajectories	Bimanual collection at scale; hardware count is not embodiment count ^[132]
UMI / DexUMI	Handheld gripper or human-hand interfaces	Portable demonstrations still need temporal, pose, and reachability alignment ^{[130][147]}
RoboCasa365	365 tasks, 2500+ kitchens; 600+ human and 1600+ synthetic demonstration hours	Compositional and scene coverage; not all hours are real-robot collection ^{[149][154]}

These quantities cannot be added into a total of independent embodied data. Trajectories recur across datasets and training mixtures; frames, steps, clips, episodes, and hours are different units. Human video supplies objects, tasks, and motion priors but usually lacks actuator commands and touch. Real trajectories ground actions but are expensive to diversify. Simulation provides rewards and difficult scenarios while retaining contact and sensor gaps. Data value should be decomposed by scenes, embodiments, skills, recovery, and sensing modalities.

LeRobot provides dataset and robot interfaces, while Isaac Lab and ManiSkill provide learning and simulation infrastructure. Standard containers do not standardize control semantics. Joint ordering, rotation conventions, normalization, timestamps, calibration, and control periods still require explicit alignment. ^{[138][140][141]}

Benchmarks ask different questions

Benchmark	Main target	Conditions that must match
LIBERO	Language manipulation and spatial/object/goal transfer	Suites, preprocessing, demonstration count, resets, rollout horizon ^[134]
CALVIN	Sequences of language-conditioned subtasks	Environment split such as ABC-D; mean chain length is not success percentage ^[135]
SIMPLER	Simulation proxies for particular real setups	Visual Matching versus Variant Aggregation, robot, control frequency ^[136]
RoboTwin 2.0	Bimanual manipulation, randomization, embodiment variation	Clean/randomized conditions, tasks, action interfaces ^[137]
RMBench	Manipulation requiring historical information	Memory split, history length, checkpoint revision; retrained weights appeared in 2026 ^[150]
RoboCasa365	Composed kitchen skills and environment transfer	Version, task level, timeout; horizons increased 1.5x in May 2026 ^[149]
BEHAVIOR-1K	Long household activities and interaction states	Challenge subset, observations, timeout, embodiment; not all 1000 activities solved ^{[139][155]}
MolmoSpaces	Multiple evaluation settings and scene coverage	Training data, action space, task coverage, adaptation, test date ^[146]

There is consequently no overall ranking of every model. High LIBERO performance after task-specific training and lower zero-shot performance in new scenes address different difficulties. A new model can also arrive with longer horizons, more sampled candidates, or larger images. Meaningful comparisons report training exposure to tasks, objects, scenes, and embodiments, as well as checkpoint version, control period, inference budget, and trial count.

A proposed evaluation protocol

The following is a proposed research and engineering design, not an experiment performed in this repository. Use four levels: in-distribution skills; new objects, views, and language; new scenes and task compositions; and new embodiments plus sustained autonomous operation. Include disturbances, occlusions, failed grasps, and recovery at every level. Separate train, validation, and test data by scenes and collection sessions and document potential pretraining overlap.

A first baseline group holds action interface and data budget constant while comparing ACT/DP, adaptable VLAs, and structured-planning systems. A second fixes the backbone and varies single-step versus chunks, discrete versus flow generation, sensing, synchronous versus asynchronous execution, and memory. A third compares world models without search, with fixed-budget search, and with increased-budget search. When commercial training data cannot be controlled, interpret results as system acceptance tests rather than causal evidence about architecture.

Record task and complete-chain success, completion time, interventions per hour, recovery, collisions/drops/stops, median and tail observation-to-action latency, energy per task, and data/adaptation cost. Prespecify repetitions per task and condition and report confidence intervals. Stratify by task and scene rather than treating correlated clips as independent trials. Every attempt, including failures and timeouts, belongs in the log.

8. A synthesis of model relationships

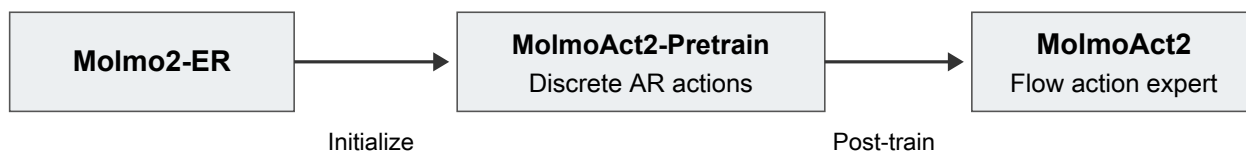
Learning paths across VLA and world models

The label on each connection states what is transferred or changed.

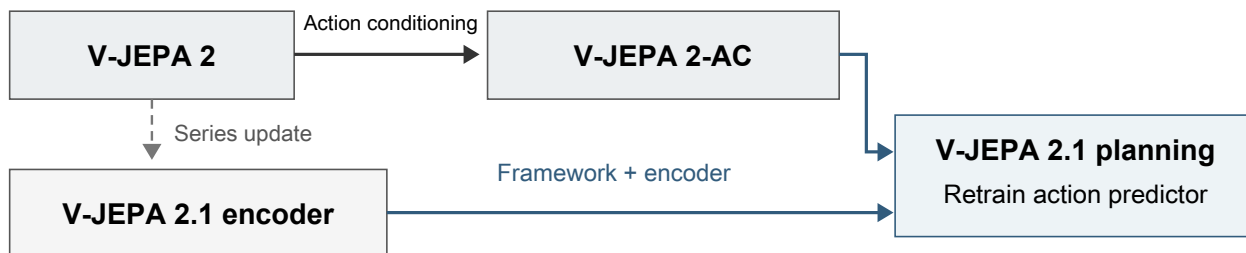
1. VLA adaptation and action-output recipe



2. VLM initialization, discrete pretraining, continuous actions



3. Action conditioning and an updated visual representation



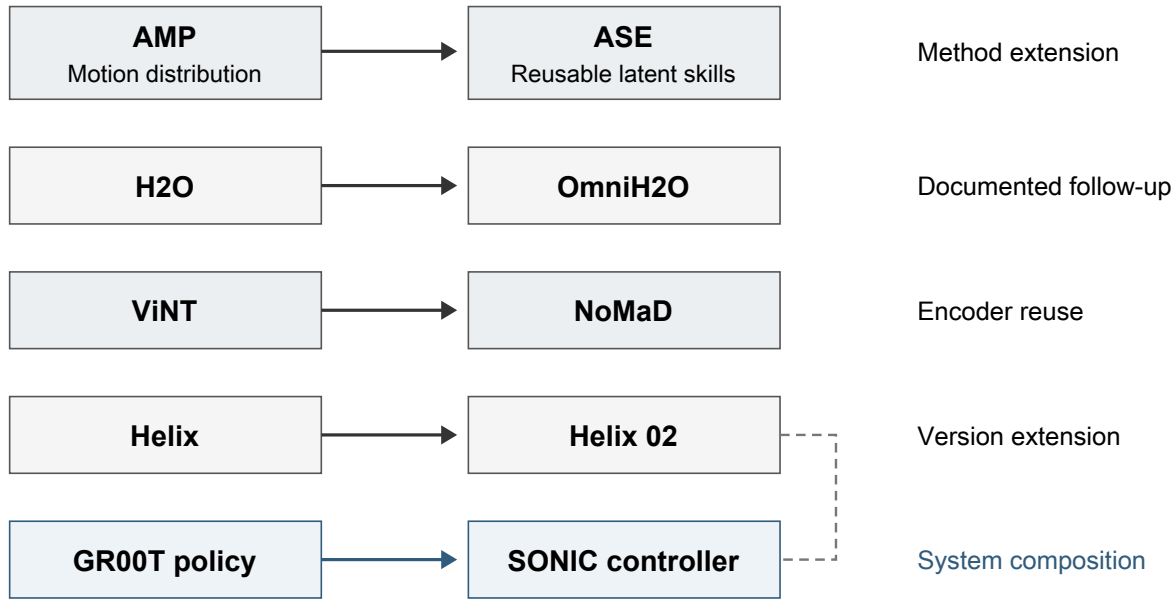
Solid: specified method path. Blue: components combined for planning.

Dashed: encoder-series progression, not inheritance of action-predictor weights.

Training paths and framework reuse

Model relationships: lineage and composition

A family progression does not imply shared weights.



Solid: documented relationship. Blue: linked execution components.

Dashed: hierarchical-control analogy between Helix 02 and SONIC, not ancestry.

Model relationship map

Relationships are not one lineage

The relationship inventory distinguishes **explicit initialization/reuse, series evolution, method combination, system complementarity, and analytical comparison**. Weight inheritance or post-training on another model is claimed only when an original paper or official implementation says so. Sharing a transformer or flow objective establishes a common paradigm, not shared weights. Each recorded edge has evidence and a relationship type.

The first connection runs from **specialized imitation to general conditional policies**. ACT and DP address sequence modeling and multimodal control; RT, Octo, and OpenVLA emphasize broader task and embodiment coverage. Generalist models do not invalidate compact imitation baselines in fixed tasks with limited demonstrations or compute. Large VLAs also adopt chunks and generative heads. Matched task data are required to isolate the benefit of internet pretraining.

The second is **a more explicit division between semantics and action distributions**. RT-2 maps actions into a language-style token space; OpenVLA provides an open transfer route. PI, GR00T, MolmoAct2, and Xiaomi use continuous action experts to address fine control and long action sequences. FAST changes encoding, OFT changes adaptation and output recipes, and asynchrony changes scheduling. Representation, training, and runtime should not be collapsed into a succession of model generations.

The third is **complementarity between geometry and scale**. PerAct and RVT align scene and action geometry, while 3D Diffuser Actor adds generative action modeling. Strong semantic pretraining does not guarantee precise coordinates, collision boundaries, or contact states. Geometry can provide an inspectable interface at a calibration and compute cost. The source of spatial error should determine whether to combine these components.

The fourth is **convergence between world models and VLAs without identical roles**. GR-1/GR-2 use future prediction in action learning; GO-1 uses latent actions as an intermediate plan; Cosmos Policy adapts a video foundation model to actions, futures, and values; JEPA-AC searches through latent action consequences. These correspond to auxiliary supervision, intermediate representation, unified generation, and explicit planning. Deployment-time world-model planning should be claimed only when inference actually evaluates possible consequences. ^{[132][133][75][76][83][90][78][80][81]}

The fifth is **integration of general reasoning with low-level whole-body control**. HumanPlus, SONIC, and Helix illustrate separate timescales for semantics, motion targets, and stabilization. A VLA emitting a whole-body joint vector has not automatically solved balance and contact; a stable locomotion controller does not automatically understand open-ended tasks. Their interface needs coordinates, timing, constraints, and failure feedback.

The sixth is **a feedback cycle connecting imitation and policy improvement**. PI's experience-learning work and HIL-SERL exemplify broad pretrained adaptation and bounded-task feedback optimization. Both use outcomes to improve behavior, but rewards, scale, released assets, and operating constraints differ. World models can reduce physical exploration cost while introducing errors that an optimizing policy may exploit. ^{[142][143]}

Six useful direct comparisons

Comparison	Actual distinction	Appropriate test
ACT versus DP	Latent-variable chunk prediction versus action denoising	Matched demonstrations and feedback interval: precision, failure, latency
OpenVLA versus OFT	Adaptation and output recipe around a reusable backbone	Fixed backbone and training budget
PI versus GR00T	Related continuous-expert paradigm with different backbones, mixtures, embodiments, versions	Pinned checkpoints and permissions; embodiment adaptation cost
V-JEPA-AC versus video world models	Compact latent consequences versus viewable futures	Fixed search time/candidates and real task success
Planner plus skills versus end-to-end VLA	Explicit subgoals versus a learned joint action interface	New compositions, failure recovery, grounding, module overhead
Helix/SONIC hierarchy versus arm-only VLA	Whole-body balance, contact frequency, and stabilization	Manipulation while moving, disturbances, multiple contacts

9. Engineering architecture and openness

A practical architecture separates data/calibration, state/history, task planning, action generation, control constraints, and feedback logging. Slower semantic inference may run on a server, an action expert near the robot, and servo control and stopping on deterministic control paths. Frequencies must follow hardware and measured requirements rather than System 1/System 2 terminology. Every executed command should be traceable to the observation, model, and normalization revision that produced it.

Audit openness along five axes: disclosed mechanism, inference code, weights, training code/recipe, and actual training data, then separately record license and use restrictions. A GitHub repository does not establish completeness along all five axes. Hardware recommendations do not prove real-time performance in every mode. Public data and intermediate checkpoints facilitate reproduction, but calibration, controllers, and recovery can still depend on unrecorded operational expertise. ^{[140][141][144]}

Selection should start from task constraints. A fixed precision workstation can compare ACT/DP and HIL-SERL; language generalization and task sharing motivate adaptable VLAs; explicit spatial constraints motivate geometry; choices among consequences motivate world-model planning. For mobile manipulation, establish low-level stability before connecting general semantics. These are mechanism-based recommendations, not procurement rankings or deployment guarantees.

10. Five future technical challenges

Challenge 1: transferable action and contact representations

Sharing semantics across embodiments is easier than sharing actuator meaning. Joints, grippers, hands, control rates, and force sensing change the action distribution, while human video lacks command labels. Research should encode embodiment constraints, geometric goals, and contact states explicitly and use adaptation layers beneath shared policies. Tests should include new-embodiment demonstration requirements, contact-force error, insertion success, constraint violations, and post-transfer stability rather than source-robot success alone.

Challenge 2: persistent memory, verifiable plans, and recovery

Tasks require remembering hidden objects, completed steps, and failed strategies. More video history increases cost and can preserve irrelevant context, while text plans can drift from actual state. Event memory, object-state graphs, explicit completion predicates, and recovery branches offer testable directions. RMBench and long-horizon benchmarks provide partial entry points, supplemented by real occlusion, moved objects, and interruptions. Report complete-chain success, recovery time, and persistence of incorrect memories.^{[150][139]}

Challenge 3: world models that support action decisions

A visually plausible future can be wrong for control, especially under contact, deformation, occlusion, or interaction with others. Longer planning amplifies error and can find simulator loopholes. Priorities include calibrated action conditioning, uncertainty, physical constraints, and correction from real feedback. Evaluation needs counterfactual action discrimination, agreement between predicted and real outcomes, fixed-budget planning gains, and conservative fallback under unexpected conditions.

Challenge 4: jointly optimizing latency, energy, and control reliability

Large visual-language backbones, long histories, and iterative action generation compete for on-device resources, while buffering changes feedback age. Quantization, distillation, KV reuse, chunk scheduling, and control stability should be designed together and tested under disturbances. Tail latency, observation age, continuity, energy per task, and recovery matter alongside throughput. Xiaomi's asynchronous training provides a concrete reason to align training conditions with scheduling.^{[151][152]}

Challenge 5: auditable continual improvement beyond demonstrations

Selecting successful videos, reporting the best checkpoint, or hiding interventions overstates operational capability. Continual learning adds reward mismatch, forgetting, leakage, and rollback problems. Preserve all attempts, interventions, and failures, use independent-scene tests, and support reversible model releases. Track licenses for videos, scene assets, weights, and training data separately. Regression across versions, confidence intervals, human minutes, and sustained intervention rates reveal whether systems actually reduce human work.^{[142][146][149]}

11. Reading and reproduction path

A useful progression is ACT/DP for action learning, SayCan/PaLM-E for grounding, RT/OpenVLA for transfer, PI/GR00T for continuous experts, PerAct/RVT for geometry, followed by JEPA/Cosmos and whole-body control. Reproduction should begin with a fixed robot, explicit action interface, and repeatable tasks before adding out-of-distribution changes and recovery. Simultaneously replacing sensors, embodiment, data, and model prevents clean causal interpretation.

The repository retains model cards, relationship edges, data and benchmark tables, a product matrix, evidence and uncertainty records, both report editions, and build/validation scripts. It provides literature analysis and an implementable evaluation design, not reproduced training results, third-party paper copies, weights, or raw robot data. Updates should record version changes in the source ledger and revalidate agreement between diagrams, tables, and prose.

12. Model and method card index

Entries include models, versions, methods, and components, not that many independent foundation models. Full field-by-field decompositions are in the [Model Atlas](#).

ID	Model / method	Year	Evidence
act	ACT	2023	[1][2]
diffusion-policy	Diffusion Policy	2023	[3][4]
rt1	RT-1	2022	[5][6]
rt2	RT-2	2023	[7][8]
rtx	RT-X / Open X-Embodiment	2023	[9][10]
octo	Octo	2024	[11][12]
openvla	OpenVLA	2024	[13][14]
openvla-oft	OpenVLA-OFT	2025	[15][16]
pi0	pi0	2024	[17][18][20]
fast	FAST / pi0-FAST	2025	[19][20]
pi05	pi0.5 / knowledge insulation	2025	[21][22][20]
pi06	pi0.6	2025	[23][24]
pi06-recap	pi-star-0.6 / RECAP	2025	[24][25]
mem	MEM / pi0.6-MEM	2026	[28]
pi07	pi0.7	2026	[26][27][20]
smolvla	SmoIVLA	2025	[29][30]
groot-n1	GR00T N1	2025	[31]
groot-n15	GR00T N1.5	2025	[32][35]
groot-n16	GR00T N1.6	2025	[33][35]

ID	Model / method	Year	Evidence
groot-n17	GR00T N1.7	2026	[34][35]
gemini-robotics1	Gemini Robotics / ER	2025	[36][37]
gemini-robotics15	Gemini Robotics 1.5 / ER 1.5	2025	[38][39]
gemini-er16	Gemini Robotics-ER 1.6	2026	[40][41]
gemini-robotics2	Gemini Robotics 2	2026	[42]
gemini-er2	Gemini Robotics ER 2	2026	[42][43]
gemini-ondevice2	Gemini Robotics On-Device 2	2026	[42][44]
rdt1b	RDT-1B	2024	[45][46]
univla-odl	UniVLA (OpenDriveLab)	2025	[47][48]
univla-baai	UniVLA (BAAI)	2025	[49][50]
xvla	X-VLA	2025	[51][52]
lingbot-vla2	LingBot-VLA 2.0	2026	[53][54]
rtc	Real-Time Chunking (RTC)	2025	[55]
saycan	SayCan	2022	[56]
palme	PaLM-E	2023	[57]
embodiedgpt	EmbodiedGPT	2023	[59]
vima	VIMA	2023	[58]
voxposer	VoxPoser	2023	[60]
r3m	R3M	2022	[61]
vc1	VC-1	2023	[62]
peract	PerAct	2022	[63]
rvt	RVT	2023	[64]
3d-diffuser-actor	3D Diffuser Actor	2024	[65][66]
dreamerv3	DreamerV3	2023	[67][68]
unisim	UniSim	2023	[69][70]
genie1	Genie	2024	[71]
genie2	Genie 2	2024	[72]
genie3	Genie 3	2025	[73][74]
gr1	GR-1	2023	[75][76]
gr2	GR-2	2024	[77][91]

ID	Model / method	Year	Evidence
vjepa2	V-JEPA 2	2025	[78][79][81]
vjepa2-ac	V-JEPA 2-AC	2025	[78][79][81]
vjepa21	V-JEPA 2.1	2026	[80][81]
cosmos-predict25	Cosmos Predict 2.5	2025	[82]
cosmos-policy	Cosmos Policy	2026	[83][90]
cosmos3	Cosmos 3	2026	[84][85]
gaia2	GAIA-2	2025	[86]
gaia3	GAIA-3	2025	[87]
gaia4	GAIA-4	2026	[88]
cosmos-predict2	Cosmos Predict2-2B	2025	[83][90]
H-M01	DeepMimic	2018	[92]
H-M02	AMP / ASE	2021	[93][94]
H-M03	PHC	2023	[95][96]
H-M04	Humanoid Transformer	2023	[97]
H-M05	H2O / OmniH2O	2024	[98][99]
H-M06	HumanPlus	2024	[100]
H-M07	MaskedMimic	2024	[101]
H-M08	SONIC / GEAR-SONIC	2025	[102][103][123]
H-M09	ViNT	2023	[104][105]
H-M10	NoMaD	2023	[106][105]
H-M11	GOAT	2023	[107][108]
H-M12	Helix / Helix 02	2025	[109][110]
H-M13	Skild Brain / S1	2025	[111][112][113]
go1	GO-1 / GO-1 Air	2025	[132][133]
serl	SERL	2024	[143]
hil-serl	HIL-SERL	2024	[142][153]
molmoact2	MolmoAct2	2026	[144][145]
xiaomi-robotics-0	Xiaomi-Robotics-0	2026	[151][152]

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